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Prevalence of hepatitis B virus and associated factors among blood donors in Hossana blood bank catchment area, Southern Ethiopia

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Abstract

Background Human hepatitis is an inflammation of the liver brought on by the DNA virus known as the hepatitis B virus (HBV). Around the world, 240 million people are thought to have HBV in a chronic state. The prevalence of viral hepatitis is extremely high in Africa. Prior reports from various regions of the nation have indicated a variable prevalence of HBV infection.

Objective To estimate the prevalence of HBV infections and associated factors among blood donors in the Hossana blood bank catchment area in August 2023.

Methods A community-based cross-sectional study design was conducted among 546 blood donors in the Hossana blood bank catchment area from August 22 to September 22, 2023. Study participants were included using the system established by the blood bank. Both Binary and multivariable logistic regression analysis was conducted. The association of variables was declared at p-value < 0.05 and strength of association was presented using AOR with a corresponding 95% CI. The adequacy of the model was checked by Hosmer-Lemeshow test.

Results a total of 546 participant's data was analyzed. The prevalence of HBV among blood donors in the Hossana blood bank catchment area was 7.88 with 95% CI [5.76–10.46]. No participants received post-donation counseling. Having multiple sexual partners (AOR = 5.18, 95% CI: 1.32–20.36), Sharing of sharp material (AOR = 3.66, 95% CI: 1.32–10.13) and tooth extraction (AOR = 5.55, 95% CI: 2.08–14.78) were identified as factors associated with HBV positivity.

Conclusion In this study, the prevalence of hepatitis B virus was 7.88, which is intermediate. Multiple sexual partners, sharing of sharp material, and tooth extraction were related to the hepatitis b virus. Though post-donation counseling is important for the control and prevention of hepatitis b virus infection, it has been neglected since the establishment of the blood bank we strongly recommend counseling regular donors.

Clinical trial number Not applicable.

Keywords Prevalence, HBV, Factors, Blood donors

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Background

Human hepatitis is an inflammation of the liver brought on by the DNA virus known as the hepatitis B virus (HBV) [1]. Based on the evolutionary study, eight genotypes for HBV have been confirmed (labeled A to H), two tentative genotypes (labeled I and J), and close to forty sub-genotypes [2]. These eight different genotypes of HBV have been identified by phylogenetic and evolutionary analysis of full genomic sequences; each has a larger than 7.5% intergroup nucleotide difference [3].

The virus can spread horizontally during unprotected sex, vertically at birth, by sharing injecting equipment, and by intimate contact with newborns and babies. Since blood stays infectious even after drying for several weeks, transmission by unscreened blood products is another method of transmission [4]. HBV is present in a variety of bodily fluids. As a result, skin fractures, medicine injections administered intravenously, and the transfusion of blood and blood products can all spread it [5]. Hepatitis viruses can cause acute or chronic hepatitis that can result in death, severe liver scarring (cirrhosis), liver failure, and liver cancer [4].

It is a serious public health issue all around the world: a third of the world's population has HBV infection, and 360 million of these people are chronic carriers [2]. According to estimates from the World Health Organization (WHO) and the US (United States) Centers for Disease Control (CDC), more than 500 million people worldwide today have chronic viral hepatitis. The prevalence of viral hepatitis is extremely high in Africa [6]. Regarding its endemicity, hepatitis infection is classified as high when it is above 8%, as intermediate when it is between 2% and 7%, and as low endemicity when it is below 2% [7].

The majority of nations are still regarded as having intermediate to high HBV infection endemicity with an HBV prevalence rate of above 8% [7]. Africa is a high endemic region with the second-highest number of chronic carriers after Asia [8]. Ethiopia has been listed as one of the regions with viral hepatitis that is extremely endemic, along with the majority of African nations [8]. According to recent studies, Ethiopia has a high prevalence of HBV genotypes A, D, C, E, and G [9].

Infection with the hepatitis B virus affects public health and is a leading cause of illness and mortality, especially in poor nations. It causes about one million deaths annually, 57% of liver cirrhosis cases, and 78% of instances of primary liver cancer across the globe. Hepatocellular carcinoma (HCC), the most common form of primary liver cancer, is caused by chronic hepatitis B in 60–80% of cases worldwide [10, 11]. Moreover, chronic HBV infections are the cause of up to 80% of hepatocellular malignancies and more than 60% of chronic liver disorders in Kenya and Ethiopia, respectively [12].

Gathering the most recent data on HBV prevalence in Ethiopia could be essential for putting into practice intervention strategies and recommendations to lessen this global and national public health issue. To the best search of the literature, the precise prevalence of these viruses in Ethiopia is still unknown because there has not been a comprehensive study and there are no easily accessible routine serological tests for patients. Even though there have been a few studies, they have typically either been limited to people in risk categories, HIV patients or had a tiny sample size [13, 14].

Those limited previous studies among blood donors in Ethiopia have concentrated on all types of donors, including commercial blood donors and familial replacement blood donors, but the focus of this study is to estimate prevalence only among volunteer blood donors and further identifies the potential factors including some variables that are not studied by previous studies like previous history of post-donation counseling. Moreover, although volunteer donors play a paramount importance in enhancing the blood donation scheme, the factor of post-blood donation counseling for regular donors is not well addressed and studied by the prior studies. Likewise, the prior studies tried to assess the problem under study from the health care provider's point of view rather than volunteer regular donors. In addition, those existing studies assessed the problem under study with a relatively low sample size, which may underestimate the problem and hide the potential contributing factors.

This study, therefore, determines the prevalence of HBV and associated factors in willing blood donors from the Hossana blood bank catchment area. This will aid in the prevention and control of HBV infection and guarantee blood safety in pursuing national blood transfusion services, strategies, and improvement of the blood donor selection protocol by geographic area and population subgroup. Finding the causes is also necessary for creating specialized interventions to protect blood donors from HBV exposure and transmission.

Method and materials

Study setting and period

The study was conducted in the Hossana blood bank catchment area from August 22 to September 22, 2023. The total population of the catchment area was 75,963. The annual capacity of blood collection is around 10,000 units. The expected number of volunteer blood donors is 14,000 annually.

Study Design

A community-based cross-sectional study was conducted.

Population

All adults aged between 18 and 65, and who volunteered to donate in the Hossana blood bank catchment area during the given study period were the study population.

Eligibility criteria

Inclusion criteria

All adults who are volunteer blood donors in the Hossana blood bank catchment area.

Exclusion criteria

Participants who encountered faint were excluded.
Participants with insufficient sample.
Participants (donors) Repeated code.
Participant with unlabeled blood collection bag.

Sample size determination

The sample size for the prevalence was calculated by using a single population proportion formula considering the following assumption taking HBV proportion (p) of 6.2% from literature conducted in Amhara and Tigray region blood donors and a 3% margin of error is used [15].

$$n = \quad Z^{2}_{\frac{\alpha}{2}} \frac{p(1-p)}{d^2}$$

n = required sample size, Z = 1.96 at 95% confidence level.
d = margin of error, p = population proportion.

$$n = \frac{(1.96)^2 * 0.062 * (1-0.062)}{(0.03)^2} = 248$$

By adding a non-response rate of 10% = 273

The sample size for the factors was determined by using open epi info version 3 by considering the following assumptions; confidence interval of 95%, power = 80%, ratio of exposed to non-exposed as 1:1 and summarized in the (Table 1).

Based on the above result, the sample size calculated for factors is 462, which is greater than the first specific objectives, and then by adding a 10% non-response rate the final sample size is **508**.

Sampling procedures

Through the system established by the blood bank, the mobilization of various community segments (secondary school students, university students, religious organizations, and youth centers) was done. Volunteers who visit the blood bank were included in this study. To collect blood from willing blood donors, three functional teams of blood bank workers were developed. Depending on the number of volunteers at each place, blood was collected. Each team aimed to collect 110 units of blood over 15 days at the blood bank center and fifteen days of outreach. The calculated sample size, which is 508, is not much less than the expected number of blood donors, which is 660, in the given study period. Therefore, the study covered all volunteer blood donors who contributed to the Hossana blood bank between August 22, 2023, and September 22, 2023, including the total sample number of volunteers who donated blood only once throughout the study period. Assuming the presentation of donors to the collection sites is random, all volunteer blood donors undergo a thorough history check, physical examination, and screening to determine their eligibility for donation. Finally, those participants fulfilling the inclusion criteria were included and interviewed.

Study variables

Dependent variable

Hepatitis B infection status.

Independent variables

Socio-demographic Sex, Age, Income, Marital status, Residence, Ethnicity, Religion and Educational level.

Clinical factors IVDU, Frequency of blood donation, Prior history of blood donation, Interaction with jaundiced patient, Tooth extraction, History of blood transfusion, Hospitalization, HBV vaccine status, History of STI, and Prior history of post-donation counseling.

Behavioral factors Number of sexual partners, History of exchanging sharp material, History of smoking, History of alcohol consumption, and History of tattooing.

Operational definition

Hepatitis B Surface antigen (HBsAg) is an antigen found on the virus's exterior. Its existence suggests HBV infection [18].

Table 1 Showing sample size calculation for associated factors in the Hossana blood bank catchment area, 2024

Variables	CI	Power	Ratio of unexposed to exposed in sample	Percent of unexposed with HBV	OR	Sample size(*n*)	Ref
Age (18–24)	95%	80	1	7.5	2.5	424	[17]
Income	95%	80	1	10	0.28	446	[17]
Occupation	95%	80	1	5	2.8	462	[16]

Hepatitis B surface antigen seropositivity; - is defined as the presence of HBsAg in serum or plasma as determined by a diagnostic kit [18].

Alcohol consumption; -is the drinking of beverages containing ethyl alcohol. According to NIAAA, heavy drinking is as follows:

Having more than four drinks on any one day or more than 14 drinks per week for men.

Having more than three drinks on any one day or more than seven drinks per week for women [19].

Smoking smoking is the practice of inhaling and exhaling burning plant material vapors. Although several plant materials are smoked, such as hashish and marijuana, tobacco is the one most frequently connected to the act when it is smoked in a cigarette, cigar, or pipe [20].

Post-donation counseling; - Post-donation counseling is a private conversation between a blood donor and a qualified counselor concerning matters, to describe the test results, the requirement for results confirmation, the health effects on the donor and the provided blood (discard), and the donor's appropriateness for future blood donations [21].

Multiple sexual partners; - is the measure and incidence of engaging in sexual activities with two or more people within a specific period. Sexual activity with multiple sexual partners can happen simultaneously or serially. Multiple sexual partners include sexual activity between people of a different gender or the same gender.

Data collection procedures and measurement

A standardized questionnaire was used from the prior literature to collect data on socio-demographic, clinically relevant, and psycho-socially linked factors. The questionnaire was initially produced in English, and then translated to respective local languages, and back into English to check the language consistency. The data gatherers obtained each participant's basic information after the donors gave their consent to take part in the study and complete an informed consent form. For every blood donor, five milliliters of venous blood were extracted using sterile test tubes. Two milliliters of serum were extracted from each sample using sterile plastic vials after the serum was separated from the cells by centrifugation at a speed of 3500 revolutions per minute (RPM) for five minutes. Using an icebox cold chain device, the collected serum was delivered to the Hawassa Blood Bank laboratory. If the analysis was not complete in 8 h, the samples were warmed to room temperature before the analysis and stored at + 2 °C to + 8 °C for a maximum of 7 days. A senior blood bank lab technician prepared the serum. Each donor's serum was examined for HBsAg using an automated alinity device at one time can test around 150 samples. Three nurses were recruited for data collection.

Data quality management

To maintain the consistency of the questions, the questionnaire was initially written in English and translated in to Amharic, and then back-translated to English by linguists. Ten days before the beginning of the actual data collection, data collectors received training on data collection and sample methodologies as well as pre-testing of the questionnaire on 5% (5%) of the blood donors was conducted in Areka town. To ensure consistency and completeness of the information obtained, investigators were evaluated and spot-check the completion of questionnaires on a daily basis. While collecting data, blood donors' privacy was maintained.

Data Processing and Analysis

The information gathered from the questionnaire was collected by using an open data kit (ODK), cleaned up, and analyzed with STATA version 14. To describe the data, frequency, percentage, and mean were employed. The data were presented via tables and figures. To find potential factors influencing the outcome, binary logistic regression with both bi-variable and multivariable options was used. All variables with p-value < 0.25 in bi-variable analysis entered into a multivariable logistic regression model to control for all possible confounders and to identify factors of the outcome variables. Adjusted odds ratio along with 95% confidence interval was estimated to measure the strength of association. The fitness of the model was tested by the Hosmer-Lemshow goodness of fit test. Finally, statistical significance was declared at p-value < 0.05.

Result

Socio-demographic characteristics of the respondent

Data from 546 blood donors in the Hossana blood bank catchment were collected. Most of the donors 426(78.02%) were female and about 256 (46.89%) of respondents were within the age category 25 to 34 years. Among the study participants, 426(94.51%) were urban resident and 344(63.00%) attended college and above followed by secondary school 182(33.33%) (Table 2).

Clinical factors of the respondents

Most of the study participants, 442 (80.95), have a prior history of blood donation. Among the study participants, 538(98.53) do not interact with jaundiced patients and almost all of the respondents have no history of blood transfusion (Table 3).

Behavioral factors of respondents

Most of the study participants 438(80.51) have one sexual partner. None of them had practiced tattooing. Among the respondents, about 70% have not ever

Table 2 Socio-demographic characteristics of respondents among blood donors of Hossana blood bank catchment area, Hossana, Ethiopia, 2023

Variable	Category	Frequency (percent)
Age	18–24	144(26.37)
	25–34	256(46.89)
	35–44	120(21.98)
	>=45	26(4.76)
Sex	Male	120(21.98)
	Female	426(78.02)
Residence	Rural	30(5.49)
	Urban	516(94.51)
Marital status	Divorced	6(1.10)
	Married	222(40.66)
	Single	318(58.24)
Educational status	Can't read and write	12(2.20)
	Can read write	4(0.73)
	Primary	4(0.73)
	Secondary	182(33.33)
	Above secondary	344(63.00)
Ethnicity	Gurage	162(29.67)
	Hadiya	90(16.48)
	Halaba	88(16.12)
	Kambata	82(15.02)
	Silte	102(18.68)
	Others	22(4.03)
Religion	Muslim	206(37.73)
	Orthodox	140(25.64)
	Protestant	194(35.53)
	Others	6(1.10)

Table 3 Clinical factors of the respondents among blood donors of Hossana blood bank catchment area, Hossana, Ethiopia, 2023

Variable	Category	Frequency (percent)
Intravenous drug use	Yes	22(4.03)
	No	524(95.97)
Prior history of blood donation	Yes	442(80.95)
	No	104(19.05)
Prior history of post-donation counselling	No	440(99.55)
	Yes	2(0.45)
Dental extraction history	Yes	62(29.67)
	No	384(70.33)
Interaction with jaundiced patients	Yes	8(1.47)
	No	538(98.53)
History of blood transfusion	Yes	4(0.73)
	No	542(99.27)
History of hospitalization	Yes	8(1.47)
	No	538(98.53)
History of STI	Yes	544(99.63)
	No	2(0.37)

Table 4 Behavioral factors of the respondents among blood donors of Hossana blood bank catchment area, Hossana, Ethiopia, 2023

Variable	Category	Frequency (percent)
Number of sexual partners (544)	No	74(13.60)
	One	438(80.51)
	Two and greater than two	32(5.88)
Smoking history	Yes	40(7.35)
	No	504(92.65)
History of alcohol consumption	Yes	38(6.99)
	No	506(93.01)
Sharing of sharp material	Yes	30(5.51)
	No	514(94.49)

drunk alcoholic drinks and 92.65% have not ever smoked tobacco products (Table 4).

The prevalence HBV

From the 546 respondents, 43 were positive for HBV. The prevalence of HBV among blood donors in Hossana blood bank catchment area was 7.88% with 95% CI interval of [5.76–10.46]. From 40 smokers and 106 respondents having more than one sexual partner tested for HBV, 55% and 34% were positive, respectively.

The risk factors associated with HBV

Based on the p-value of the Bivariable analyses, six variables were identified as candidate variables for the multivariable model. These were sharing of sharp material, alcohol consumption, smoking, multiple sexual partner, jaundice and tooth extraction. The result of the multivariable analysis identified multiple sexual partner, sharing of sharp material, and tooth extraction as independent risk factors of HBV among blood donors of Hossana blood bank catchment area. Those who have tooth extraction are 5 times more likely to be positive for HBV as compared to those who has no tooth extraction keeping all other covariates constant (Table 5).

Discussion

In this study, the prevalence of HBV among blood donors in the Hossana blood bank catchment area was 7.88. This is in line with cross-sectional studies conducted in Tigray and Amhara region [22], Hossana town [23] and Jigjiga [24]. On the other hand, the finding was greater than the study conducted in Arbaminch [15], Hawassa [25], Bahir Dar [26] and Dire Dawa [1]. This might be due to the actual differences in study population, duration of the study, and geographical differences of the study populations; moreover, the screening procedure may also play a role in the observed variations. Previously the screening procedure was by using ELISA. Now, the procedure was shifted to PCR, which has the capability to identify

Table 5 Bivariable and Multivariable Logistic regression analyses results of factors associated with HBV among blood donors of Hossana blood bank catchment area, Hossana, Ethiopia, 2023

Variables	Frequency	COR 95% CI	AOR 95% CI
Sharing Sharp materials			
No	514(94.49)	1	1
Yes	30(5.51)	12.90[6.08–27.35]	**3.66[1.32–10.13]***
Alcohol			
No	384(70.33)	1	1
Yes	62(29.67)	14.63[6.51–32.86]	2.96[0 0.94- 9.36]
Smoking			
No	504(92.65)	1	1
Yes	40(7.35)	12.95[5.85–28.65]	2.02[0.70–5.83]
Sexual Partner			
No	74(13.60)	1	1
One	438(80.51)	18.98[11.42–27.13]	5.30[0.9-10.94]
>= 2	32(5.88)	14.20[4.45–45.32]	**5.18[1.32–20.36]***
Jaundice			
No	538(98.53)	1	1
Yes	8(1.47)	4.04[0.79–20.65]	3.89[0.44–34.44]
Tooth Extraction			
No	506(93.01)	1	1
Yes	38(6.99)	8.11[3.13–11.16]	**5.55[2.08–14.78]***

recently infected blood, and it has high sensitivity. On the contrary, the finding was less than the study conducted in Wolayita Sodo [27]. The variation may be due to the difference in sample size, the research was done on all kind of blood donors including health care workers but this study was conducted only on volunteer blood donors.

This study showed that having multiple sexual partner had a significant association with positivity of HBV. Those having multiple sexual partner were 5.18 times more likely to be infected with HBV. The finding is consistent with study conducted in Bahir-Dar [23]. The reason for this finding might be because that sexual practice with multiple partner is the crucial means of acquiring transfusion transmissible infections among blood donors. the finding is lower where compared to studies conducted in eastern Ethiopia [28].

Sharing of sharp material was significantly associated with hepatitis B virus. Those having shared any sharp material were 3.66 times more likely to be infected with HBV than who never share sharp material. Consistent report was reported from Debre Markos [19] and systematic review and meta-analysis conducted in Ethiopia [26].

Tooth extraction was significantly associated with hepatitis B virus where blood donors who had a history of tooth extraction were at high risk of hepatitis B virus. Consistent result was reported from Debre Markos blood bank [27]. A lower result was reported from Kality prison [11].

Conclusion

In this study, the prevalence of hepatitis B virus is at an intermediate level according to WHO guidelines on testing hepatitis B and C virus which states that regarding its endemicity, hepatitis infection is classified as high when it is above 8%, as intermediate when it is between 2% and 8%, and as low endemicity when it is below 2%. Multiple sexual partners, sharing of sharp material, and tooth extraction were factors associated with Hepatitis B virus. These illnesses were shown to be common among the willing blood donors in this study, even though the subjects were voluntary and seemed healthy. For this reason, it is strongly advised to strictly select blood donors using accepted procedures to guarantee the recipient's blood's safety. Though post-donation counseling is important for the control and prevention of the hepatitis b virus it has been neglected since the establishment of the blood bank we strongly recommend counseling regular donors.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12879-025-10550-0.

Supplementary Material 1: Questionnaire.

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Author contributions

The conception of the work, design of the work, acquisition of data, analysis, and interpretation of data was done by M.M. Data curation, drafting the article, revising it critically for intellectual content, validation, and final approval of the version to be published was done by M.W., E.S. All authors read and approved the final manuscript.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethical approval

Ethical approval for this study was obtained from the Wachemo University College of Medicine and Health Science IRB (Ref No/WCU/SGS 571/15, Date 24/01/2023 G.C) and delivered to the Hossana Blood Bank Administration. By employing an anonymous data-gathering platform and providing data collectors with training, we can ensure that the information we acquire about respondents remains private. Participants in the study were provide their informed permission after being instructed about information confidentiality. Participants had the option to leave the study at any time if they so desired. The thoughts and decisions of individuals were respected in order to preserve their autonomy. At the end of data collection, those volunteer blood donors who are sero-positive for HBV were informed, counselled and linked for continuum of care at health institution. All the procedures were conducted based on the principles of the Helsinki Declaration.

Consent for publication
Not Applicable.

Competing interests
The authors declare no competing interests.

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